Artificial Intelligence (AI)

Artificial Intelligence (AI) is the field of computer science that enables machines to perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, decision-making, and understanding natural language.

Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence that allows computers to learn patterns from data without being explicitly programmed. Machine learning algorithms improve their performance as they are exposed to more data.

There are three main types of Machine Learning:

1. Supervised Learning

- Uses labeled datasets.
- Examples include Linear Regression, Logistic Regression, Decision Trees, and Random Forest.

2. Unsupervised Learning

- Uses unlabeled datasets.
- Examples include K-Means Clustering and Principal Component Analysis (PCA).

3. Reinforcement Learning

- Learns through interaction with an environment using rewards and penalties.
- Commonly used in robotics and game playing.

Deep Learning

Deep Learning is a specialized branch of Machine Learning based on Artificial Neural Networks. It uses multiple hidden layers to automatically learn complex features from data.

Applications of Deep Learning include:

- Image Classification
- Speech Recognition
- Autonomous Vehicles
- Medical Diagnosis
- Face Recognition

Natural Language Processing (NLP)

Natural Language Processing enables computers to understand, analyze, and generate human language.

Common NLP applications include:

- Chatbots
- Language Translation
- Text Summarization
- Sentiment Analysis
- Question Answering

Large Language Models (LLMs)

Large Language Models are AI models trained on massive amounts of text data. They can generate human-like responses, summarize documents, answer questions, and assist with programming.

Examples of LLMs include:

- GPT-4.1
- Mistral
- Llama
- Claude

- Gemini

Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation is a technique that combines information retrieval with Large Language Models.

Instead of relying only on the model's internal knowledge, RAG retrieves relevant documents from an external knowledge base before generating an answer.

A typical RAG pipeline consists of:

1. Document Loader
2. Text Splitter
3. Embedding Model
4. Vector Database
5. Retriever
6. Large Language Model

Benefits of RAG:

- Reduces hallucinations
- Improves factual accuracy
- Uses private documents
- Provides up-to-date information
- Gives context-aware answers

Embeddings

Embeddings are numerical vector representations of text. They capture the semantic meaning of words, sentences, or documents so that similar pieces of text have similar vector representations.

Popular embedding models include:

- OpenAI text-embedding-3-small
- sentence-transformers/all-MiniLM-L6-v2
- BAAI/bge-small-en

Vector Databases

A vector database stores embeddings and performs similarity search.

Popular vector databases include:

- ChromaDB
- FAISS
- Pinecone
- Weaviate
- Milvus

Similarity search is commonly performed using cosine similarity.

LangChain

LangChain is a framework for building LLM-powered applications.

It provides components for:

- Prompt Templates
- Document Loaders
- Text Splitters
- Embeddings
- Vector Stores

- Retrievers
- Chains
- Agents
- Memory

OpenAI

OpenAI provides APIs for powerful language models and embedding models. Popular models include GPT-4.1, GPT-4.1-mini, and text-embedding-3-small.

Mistral AI

Mistral AI develops open and commercial large language models. Mistral models are known for their speed, efficiency, and strong reasoning capabilities.

ChromaDB

ChromaDB is an open-source vector database designed specifically for AI applications. It stores document embeddings and allows efficient semantic search using similarity matching.

Conclusion

Artificial Intelligence continues to transform industries by enabling intelligent systems capable of understanding language, analyzing data, and making decisions. Technologies like Machine Learning, Deep Learning, Large Language Models, and Retrieval-Augmented Generation are making AI systems more accurate, efficient, and reliable.